

CS406

Programming Practice (Java)

*Submitted partial fulfillment of the requirements for
the award of the degree of*

BACHELOR OF TECHNOLOGY

In

Computer Science & Engineering

Submitted to



**RAJIV GANDHI PROUDYOGIKI VISHWAVIDYALAYA,
BHOPAL (M.P.)**



**Department of Computer Science & Engineering
Sri Aurobindo Institute of Technology, Indore (M.P.)
2025-2026**

PROGRAMMING PRACTICE - JAVA



*The Programming Practice report submitted
to
Rajiv Gandhi Proudyogiki Vishwavidyalaya,
Bhopal, towards partial fulfillment of the
Degree of
Bachelor of Technology
in
Computer Science & Engineering*

Guided By

Prof. Gaurav Bairagi

Submitted By

Harsh kushwah
0873CS241068

**Department of Computer Science & Engineering
Sri Aurobindo Institute of Technology, Indore (M.P.)
2025-2026**

SRI AUROBINDO INSTITUTE OF TECHNOLOGY, INDORE



RECOMMENDATION

This is to certify that Mr. Harsh Kushwah (0873CS241068) is a student of B.Tech. IV Semester in the year 2025-26 of the Computer Science & Engineering Department of this institute has completed their work on “*Programming Practice*” for Java based on the syllabus and has submitted a satisfactory account of their work in this report, which is recommended for the partial fulfillment of the degree of Bachelor of Technology in Computer Science & Engineering.

Project Guide,
CSE Department
SAIT, Indore

Table of Contents

Experiment	Page No.
FRONR PAGE	I
RECOMMENDATION	II
TABLE OF CONTANTS	III
FIGURE INDEX	IV
EXPERIMENT 1: INSTALLATION OF J2SDK.	1
EXPERIMENT 2: SCOPE OF VARIABLES	1-3
EXPERIMENT 3: CONCEPT OF CLASS IN JAVA	3-5
EXPERIMENT 4: TYPE CASTING IN JAVA	5-7
EXPERIMENT 5: EXCEPTION HENDLING IN JAVA	7-9
EXPERIMENT 6: INHERITANCE	9-11
EXPERIMENT 7: POLYMORPHISM	12-14
EXPERIMENT 8: ACCESS SPECIFIERS IN JAVA	14-16
EXPERIMENT 9: USE AND ADVANTAGES OF CONSTRUCTOR	16-18
EXPERIMENT 10: INTERFACING BETWEEN TWO CLASSES	18-20
EXPERIMENT 11: ADD A CLASS TO A PACKAGE	20-21
EXPERIMENT 12: LIFE CYCLE OF A THREAD	21-23
EXPERIMENT 13: DEMONSTRATE AWT	24-26
EXPERIMENT 14: HIDE A CLASS	26-29
EXPERIMENT 15: DATA BASE CONNECTIVITY USING JAVA	29-31
EXPERIMENT 16: “HELLO JAVA” IN EXPLORE USING APPLET	32-33
EXPERIMENT 17: SHOW CONNECTIVITY USING JDBC	33-36
EXPERIMENT 18: DEMONSTRATE MULTITHREDING USING JAVA	36-38
EXPERIMENT 19: DEMONSTATE APPLET LIFE CYCLE	38-41
EXPERIMENT 20: DEMONSTATE CONCEPT OF SEVELET	41-43

FIGURE INDEX

Figure No.	Title	Description	Page No.
2.1	Scope of variable	A Java program used to demonstrate the scope and visibility of variables in different parts of a class and methods.	2
3.1	CLASS in JAVA	A Java class used to store and manage student information.	4
4.1	Type Casting	A Java program used to demonstrate type conversion and casting between different data types.	6
5.1	Exception Handling	A Java program used to demonstrate exception handling and error management techniques.	8
6.1	Inheritance	A Java class used to demonstrate inheritance by serving as the base class for derived classes.	10
7.1	Polymorphism	A Java program used to demonstrate polymorphism and dynamic method behavior.	12
8.1	Access Specifiers	A Java program used to demonstrate access control and visibility using different access modifiers.	15
9.1	Constructor	A Java program used to demonstrate object initialization using constructors.	17
10.1	Interfacing	A Java interface used to demonstrate abstraction and implementation of abstract methods	19
11.1	Package	A Java class used to demonstrate basic class and object concepts.	20
12.1	Life cycle of a Thread	A Java program used to demonstrate thread creation and multithreading concepts.	20
13.1	Demonstrate AWT	A Java program used to demonstrate GUI development using AWT components.	25
14.1	Hide a Class	A Java program used to demonstrate inner (nested) classes and their functionality.	27
15.1	Data Base Connectivity	A Java program used to connect to and interact with a database.	30
16.1	Explorer using Applet	“HELLO JAVA” IN EXPLORE USING APPLLET	32
17.1	Connectivity using JDBC	A Java program used to connect and interact with a database using JDBC.	34
18.1	Demonstrate multithreading	This is the simplest and most common way to create a thread in Java by extending the Thread class.	37
19.1	Demonstrate applet life cycle	Demonstrates the applet life cycle by simulating the execution of the main(), constructor, and paint() methods using Java Swing.	39